

TWO STRATEGIES FOR SHARP QUANTITATIVE SAINT-VENANT STABILITY VIA A SINGLE TORSION LEVEL SET

WORKING NOTE (PLAN 3)

ABSTRACT. We record, in full and honestly, the two strategies developed for the sharp quantitative Saint–Venant (equivalently, torsional Faber–Krahn) inequality $\mathcal{A}(\Omega)^2 \leq C(n) \delta_T(\Omega)$ through the device of a single torsion level set. Strategy A (“graph route”) is the by-hand Plan 1 idea executed with Serrin stability: pass to a near-Serrin level set, show it is a small graph, close perturbatively, transfer back. Strategy B (“integrated route”) uses Serrin *directly*: Weinberger’s P -function rigidity made quantitative, controlling the trace-free Hessian and hence the asymmetry, with no graph and no boundary regularity. We give the common skeleton with proofs, both attacks with detailed sketches, the tentacle analysis (including the precise origin of the w^2 escape floor that obstructs Strategy A for $n \geq 3$), the sharpness/norm-gap discussion, the comparison with the BDV selection principle, and an explicit checklist of every step requiring independent verification.

1. SETUP

Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be open with finite measure; after scaling $|\Omega| = \omega_n = |B_1|$. Its *torsion function* $u = u_\Omega$ is the unique minimiser of $J(v) = \frac{1}{2} \int |\nabla v|^2 - \int v$ over $H_0^1(\Omega)$, equivalently the variational solution of

$$(1) \quad -\Delta u = 1 \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega, \quad u > 0 \text{ in } \Omega.$$

The *torsional rigidity* is $T(\Omega) = \int_\Omega |\nabla u|^2 = \int_\Omega u = -2 \min J$. For the ball B_r , $u_{B_r}(x) = \frac{r^2 - |x|^2}{2n}$ and $T(B_r) = \frac{\omega_n r^{n+2}}{n(n+2)}$. The Saint–Venant inequality states $T(\Omega) \leq T(B)$ for the volume-matched ball B , with equality iff Ω is a ball. Define the scale-invariant deficit and the Fraenkel asymmetry

$$\delta_T(\Omega) := \frac{T(B) - T(\Omega)}{T(B)} \geq 0, \quad \mathcal{A}(\Omega) := \min_{x_0} \frac{|\Omega \Delta B(x_0)|}{|\Omega|}.$$

The target is the sharp quantitative inequality

$$(2) \quad \boxed{\mathcal{A}(\Omega)^2 \leq C(n) \delta_T(\Omega)} \quad (\delta_T(\Omega) \leq \delta_T^0(n)).$$

Sharpness of the exponent. For the ellipsoid $\mathcal{E}_\varepsilon = \{\sum x_i^2/a_i^2 < 1\}$ with $a_1 = 1 + \varepsilon$, $a_i = 1 - \varepsilon/(n-1)$ (volume fixed to first order), one computes $\mathcal{A}(\mathcal{E}_\varepsilon) \simeq_n \varepsilon$ while $\delta_T(\mathcal{E}_\varepsilon) \simeq_n \varepsilon^2$ (the first-order shape variation is volume/translation-neutral, so the deficit is purely quadratic). Hence $\mathcal{A}^2/\delta_T \rightarrow \text{const}$: the exponent 2 in (2) is optimal, and any proof must produce exactly the quadratic relation, not a fractional power. This is why every step below is tracked for the *linear*-in- δ_T (i.e. quadratic-in- $\mathcal{A}$) scaling.

2. THE COMMON SKELETON

Neither strategy argues on the rough set Ω . Instead one passes to a single superlevel set $E_{\hat{t}} = \{u > \hat{t}\}$ of its own torsion function, proves (2) *there*, and transfers it back. The structural reason to use a level set — the insight of Plan 3 — is:

$-\Delta u = 1$ has real-analytic coefficients, so u is real-analytic in the interior of Ω , with no hypothesis on $\partial\Omega$. By Sard's theorem, for a.e. $\hat{t}$ the level $\Sigma_{\hat{t}} = \{u = \hat{t}\}$ is a compact real-analytic hypersurface with $|\nabla u| > 0$. Boundary regularity of Ω is never used; only interior regularity of u , which is free.

2.1. The exact level-set deficit identity. By the coarea formula, $T(\Omega) = \int_{\Omega} u = \int_0^{\|u\|_{\infty}} m(t) dt$ with $m(t) = |\{u > t\}| = |E_t|$, and $-m'(t) = \int_{\Sigma_t} |\nabla u|^{-1} d\mathcal{H}^{n-1}$, while differentiating $\int_{E_t} (-\Delta u) = \int_{\Sigma_t} |\nabla u|$ gives $m(t) = \int_{\Sigma_t} |\nabla u| d\mathcal{H}^{n-1}$. Combining these with the isoperimetric inequality $P(t) \geq n\omega_n^{1/n} m(t)^{(n-1)/n}$ and Cauchy-Schwarz $(\int_{\Sigma_t} 1)^2 \leq (\int |\nabla u|^{-1})(\int |\nabla u|)$ produces, after the Kohler-Jobin/Talenti algebra, the *exact identity*

$$(3) \quad \delta_T(\Omega) \simeq_n \int_0^{\|u\|_{\infty}} (D_H(t) + D_I(t)) w(t) dt, \quad w(t) = (n^2 \omega_n^{2/n} m(t)^{1-2/n})^{-1},$$

where, on a regular level (with $P(t) = \mathcal{H}^{n-1}(\Sigma_t)$, $\bar{f}_t = m(t)/P(t)$),

$$(4) \quad D_H(t) = \left(\int_{\Sigma_t} \frac{d\mathcal{H}^{n-1}}{|\nabla u|} \right) \left(\int_{\Sigma_t} |\nabla u| d\mathcal{H}^{n-1} \right) - P(t)^2 \geq 0 \quad (\text{Cauchy-Schwarz / Serrin defect}),$$

$$(5) \quad D_I(t) = P(t)^2 - n^2 \omega_n^{2/n} m(t)^{2-2/n} \geq 0 \quad (\text{isoperimetric defect}).$$

$D_H(t) = 0$ iff $|\nabla u| \equiv \text{const}$ on Σ_t ; $D_I(t) = 0$ iff Σ_t is a sphere. The decisive exact reformulation is the *weighted-variance identity*

$$(6) \quad \int_{\Sigma_t} \frac{(|\nabla u| - \bar{f}_t)^2}{|\nabla u|} d\mathcal{H}^{n-1} = \frac{m(t)}{P(t)^2} D_H(t) =: \frac{m(t)}{P(t)^2} \mathcal{V}(t) (\dots),$$

obtained by expanding $\int (|\nabla u| - \bar{f})^2 / |\nabla u| = \int |\nabla u|^{-1} \bar{f}^2 - 2\bar{f}P + \int |\nabla u|$ and using $\bar{f}P = m$. D_H is the variance of $|\nabla u|$ on Σ_t weighted by $1/|\nabla u|$. This weighting is the protagonist of the whole note.

Remark 2.1 (How (3) arises). Let $\mu(t) = -m'(t) = \int_{\Sigma_t} |\nabla u|^{-1}$ and recall $m(t) = \int_{\Sigma_t} |\nabla u|$. The Saint-Venant functional admits the one-dimensional reduction $T(\Omega) = \int_0^{\infty} m(t) dt$, and along the torsion flow one has the differential relation $(\frac{d}{dt}) \int_{\Sigma_t} |\nabla u| = -P(t)^2 / \int |\nabla u|^{-1} + \dots$. Two losses appear when comparing with the ball flow: the *Cauchy-Schwarz* loss $P(t)^2 \leq (\int |\nabla u|^{-1})(\int |\nabla u|)$, whose defect is exactly $D_H(t)$ in (4), and the *isoperimetric* loss $P(t)^2 \geq n^2 \omega_n^{2/n} m^{2-2/n}$, whose defect is $D_I(t)$ in (5). Integrating the resulting differential inequality for the profile gap against the weight $w(t)$ (Talenti/Kohler-Jobin) and using $T(B) - T(\Omega) = \int_0^{|\Omega|} h$ collapses to (3). The point for us: δ_T is *exactly* the total of the two defects, each nonnegative, each with a clean geometric meaning; smallness of δ_T forces both small *in integrated average*.

2.2. Common Lemma 1: extraction of a good near-boundary level.

Lemma 2.2 (good-level extraction). *Fix a small absolute constant $\theta \in (0, 1)$. There exist C_n and a level $\hat{t}$ with: (i) $D_H(\hat{t}) \leq n^2 \omega_n \theta$, equivalently $\mathcal{V}(\hat{t}) \leq \theta$; (ii) $|\Omega \setminus E_{\hat{t}}| \leq (C_n/\theta) \delta_T(\Omega)$; (iii) $\hat{t}$ Sard-regular, so $\Sigma_{\hat{t}}$ is real-analytic, $|\nabla u| > 0$.*

Proof. Drop $D_I \geq 0$ in (3): $\int (D_H w) dt \leq C_n \delta_T(\Omega)$. Near the boundary $m(t) \uparrow |\Omega|$, so $w(t)$ is two-sidedly bounded by dimensional constants there — *not* because δ_T is small but because $m \rightarrow |\Omega|$. Change variables to the volume parameter $s = m(t) \in (0, |\Omega|)$, strictly decreasing and absolutely continuous; the layer is then *exactly linear*, $|\Omega \setminus E_t| = |\Omega| - s$. In s , the deficit integrand $\Phi(s) := D_H w |dt/ds|$ satisfies $\int_0^{|\Omega|} \Phi \leq C_n \delta_T$; by Markov the set $\{\Phi > \theta'\}$ has s -measure $\leq C_n \delta_T / \theta'$. Choosing the small-layer window $S_{\lambda} = [|\Omega| - \lambda, |\Omega|)$ with $\lambda = (4C_n/\theta') \delta_T$, the bad set occupies $\leq \frac{1}{2} \lambda$ of it, so $\{D_H \leq \theta\} \cap S_{\lambda}$ has positive s -measure; pull back to positive t -measure (strict monotone AC change) and intersect with the Lebesgue-null Sard set. (The naive “ t -length” phrasing is only heuristic: for a rough domain $\int |\nabla u|^{-1}$ may be unbounded, so there is no t -length lower bound; the volume variable removes this entirely.) $\square$

Checkpoint (C1). $w(t) = O(1)$ on the near-boundary range with dimensional constants; the volume-variable Markov bookkeeping. Routine.

2.3. Common Lemma 2: exact deficit transfer.

Lemma 2.3 (exact deficit transfer). *Put $\tilde{\Omega} := E_{\hat{t}}$, $v := u - \hat{t}$. Then $-\Delta v = 1$ in $\tilde{\Omega}$, $v = 0$ on $\partial\tilde{\Omega}$. v is exactly the torsion function of $\tilde{\Omega}$. With $\lambda = |\Omega \setminus \tilde{\Omega}|$, $\delta_T(\tilde{\Omega}) \leq (1 - \lambda/\omega_n)^{-1} \delta_T(\Omega) \leq 2\delta_T(\Omega)$, i.e. $\delta_T(\tilde{\Omega}) = \delta_T(\Omega) + O_n(\delta_T(\Omega)^2)$.*

Proof. Write the decreasing rearrangements $u_{\Omega}^*(s)$, $u_{\tilde{\Omega}}^*(s)$ and the ball comparison profiles $b_{\Omega}(s) = \frac{1-(s/\omega_n)^{2/n}}{2n}$, $b_{\tilde{\Omega}}(s) = \frac{\tilde{\rho}^2-(s/\omega_n)^{2/n}}{2n}$ ($\tilde{\rho} = (|\tilde{\Omega}|/\omega_n)^{1/n}$). Talenti's theorem gives the deficit-by-profile formula $T(B_D) - T(D) = \frac{1}{2} \int_0^{|D|} h_D(s) ds + \dots$ with $h_D = b_D - u_D^* \geq 0$. Because $v = u - \hat{t}$ is *exactly* the torsion function of $\tilde{\Omega}$ (an identity, not a comparison), the rearrangements satisfy $u_{\tilde{\Omega}}^*(s) = u_{\Omega}^*(s) - \hat{t}$ for $s \in [0, |\tilde{\Omega}|]$, while $b_{\tilde{\Omega}}(s) - b_{\Omega}(s) = (\tilde{\rho}^2 - 1)/(2n)$ is a *constant* — the $(s/\omega_n)^{2/n}$ terms cancel exactly. Hence $h_{\tilde{\Omega}} = h_{\Omega} + \Delta$ with $\Delta = \hat{t} + (\tilde{\rho}^2 - 1)/(2n)$ a constant; the exact identity $\hat{t} = u_{\Omega}^*(|\tilde{\Omega}|)$ together with the Talenti bound $u_{\Omega}^*(|\tilde{\Omega}|) \leq b_{\Omega}(|\tilde{\Omega}|) = (1 - \tilde{\rho}^2)/(2n)$ forces $\Delta \leq 0$. So the *unnormalised* deficit does not increase; the only enlargement is the volume renormalisation factor $(1 - \lambda/\omega_n)^{-1} = 1 + O(\lambda)$. No “deficit monotonicity under super-levelling” (a false statement, the V13 trap) is invoked: only the exact profile identity. $\square$

Checkpoint (C2). The profile-shift identity $u_{\tilde{\Omega}}^* = u_{\Omega}^* - \hat{t}$ and the sign $\Delta \leq 0$ (the $(s/\omega_n)^{2/n}$ cancellation is the load-bearing line).

2.4. Common Lemma 3: asymmetry transfer.

Lemma 2.4 (superlevel asymmetry transfer). *For measurable $\tilde{\Omega} \subset \Omega$, $\mathcal{A}(\Omega) \leq \mathcal{A}(\tilde{\Omega}) + 2|\Omega \setminus \tilde{\Omega}|/|\Omega|$.*

Proof. Let B be an optimal ball for $\tilde{\Omega}$, $|B| = |\tilde{\Omega}|$. Take the concentric $B' \supset B$ with $|B'| = |\Omega|$; then $|B' \setminus B| = |\Omega| - |\tilde{\Omega}| = \lambda$. By the triangle inequality for symmetric differences, $|\Omega \triangle B'| \leq |\Omega \triangle \tilde{\Omega}| + |\tilde{\Omega} \triangle B| + |B \triangle B'| \leq \lambda + |\tilde{\Omega} \triangle B| + \lambda$. Dividing by $|\Omega|$ and recalling $\mathcal{A}(\tilde{\Omega}) = |\tilde{\Omega} \triangle B|/|\tilde{\Omega}|$ with $|\tilde{\Omega}| \leq |\Omega|$ gives the claim with constant 2. $\square$

Composing Lemmas 2.2–2.4: if at the level one has $\mathcal{A}(E_{\hat{t}})^2 \leq c(n)\delta_T(E_{\hat{t}})$, then with $\lambda = O(\delta_T)$,

$$\mathcal{A}(\Omega)^2 \leq 2\mathcal{A}(E_{\hat{t}})^2 + 8(\lambda/|\Omega|)^2 \leq 2c(n)\delta_T(E_{\hat{t}}) + C'(n)\delta_T(\Omega)^2 \leq C_*(n)\delta_T(\Omega),$$

the squared collar $O(\delta_T^2) \ll \delta_T$ being the “ δ_T of room to spare”.

Checkpoint (C3). The constant 2 above, and the composition of volume rescalings across Lemmas 2.3–2.4 (no $O(\delta_T)$ term escaping to lower order than δ_T).

Everything thus reduces to the single at-level inequality $\mathcal{A}(E_{\hat{t}})^2 \leq c(n)\delta_T(E_{\hat{t}})$. The two strategies are two proofs of it.

3. STRATEGY A: THE GRAPH ROUTE

This is the by-hand Plan 1 idea with Serrin stability as the regulariser: exhibit a nearby domain that is a small *graph* over a sphere and whose asymmetry controls the original's, then invoke the sharp inequality in the nearly-spherical class. Here the nearby domain is $E_{\hat{t}}$ itself.

3.1. The intended argument. At $\hat{t}$ we have $D_H(\hat{t}) \leq \theta$, i.e. via (6) $|\nabla u|$ is $L^2(\Sigma_{\hat{t}})$ -close to a constant: the *approximate Serrin condition*. Serrin's theorem: if $|\nabla v| \equiv c$ on $\partial\tilde{\Omega}$ for the torsion function v of $\tilde{\Omega}$, then $\tilde{\Omega}$ is a ball. Quantitative Serrin stability (Aftalion–Busca–Reichel; Brandolini–Nitsch–Salani–Trombetti; Magnanini–Poggesi) should upgrade “ $|\nabla v|$ near constant” to “ $\partial\tilde{\Omega}$ is a small graph $\{(1 + \varphi(\xi))\xi\}$ over a sphere”. Then the sharp nearly-spherical Saint–Venant inequality (Fuglede; Brasco–De Philippis–Velichkov) gives $\mathcal{A}(E_{\hat{t}})^2 \lesssim \delta_T(E_{\hat{t}})$.

3.2. What the route structurally requires. To pass from the L^2 defect to the *pointwise/geometric* conclusion one needs, δ_T -uniformly:

- (NDg) $|\nabla u| \geq q_- > 0$ *everywhere* on $\Sigma_{\hat{t}}$: needed to parametrise $\{u = \hat{t}\}$ as a graph at all (the implicit function theorem divides by the normal derivative $|\nabla u|$; where $|\nabla u| \rightarrow 0$ the level can pinch or fold), and because the principal curvatures of $\Sigma_{\hat{t}}$ are $\sim |D^2 u|/|\nabla u|$.
- (Mod) a δ_T -uniform $C^\alpha(\Sigma_{\hat{t}})$ modulus of $|\nabla u|$, to run the surface interpolation $\|f\|_{L^\infty(\Sigma_{\hat{t}})} \lesssim \|f\|_{L^2(\Sigma_{\hat{t}})}^\vartheta [f]_{C^\alpha(\Sigma_{\hat{t}})}^{1-\vartheta}$,
 $\vartheta = \frac{2\alpha}{2\alpha+n-1}$, with $f = |\nabla u| - \bar{f}_{\hat{t}}$, $\|f\|_{L^2}^2 \simeq D_H$.

The interpolation inequality is the standard ball-covering one: at the L^∞ -point x_0 , on $\Sigma_{\hat{t}} \cap B_r(x_0)$ one has $|f| \geq \|f\|_\infty - [f]_{C^\alpha} r^\alpha$; choosing $r = (\|f\|_\infty / 2[f]_{C^\alpha})^{1/\alpha}$ and using the lower volume bound $\mathcal{H}^{n-1}(\Sigma_{\hat{t}} \cap B_r) \geq cr^{n-1}$ (a non-degeneracy of $\Sigma_{\hat{t}}$ itself!) gives the displayed estimate. Both (NDg) and (Mod) thus enter unavoidably.

3.3. The tentacle and the precise origin of the w^2 floor. This is the point flagged as “weird”. We derive the w^2 transparently.

Model. $\Omega = B_1 \cup T$, T a tube of axial length $L = O(1)$ and cross-radius $w \ll 1$ attached to ∂B_1 ; $|T| \simeq \kappa_n w^{n-1} L$.

(a) *Capping height* $t_{\text{cap}} \simeq_n w^2$. Freeze the tube axis; the cross-section is an $(n-1)$ -ball of radius w . To leading order u restricted to a cross-section solves $-\Delta' \phi = 1$, $\phi|_{\partial} = 0$, with the explicit barrier $\phi(r) = \frac{w^2 - r^2}{2(n-1)}$, $\max \phi = \frac{w^2}{2(n-1)}$. The transverse Dirichlet eigenvalue is $\lambda_1^\perp \simeq w^{-2}$, so by a Phragmén–Lindelöf/maximum-principle barrier the axial direction adds only a bounded correction: $\sup_T u \leq C_n w^2$ *independently of L* , and $\geq c_n w^2$. Hence

$$t_{\text{cap}} := \sup_T u \simeq_n w^2,$$

and for $t > t_{\text{cap}}$ the level set E_t does not meet T .

(b) *The unavoidable ball-shell, measure $\simeq_n w^2$* . A tentacle-free level needs $\hat{t} > t_{\text{cap}} \simeq w^2$, hence one discards $\{u < \hat{t}\}$. Besides T , this contains the boundary layer of the ball part: with $u_{B_1}(x) = \frac{1-|x|^2}{2n}$,

$$\{x \in B_1 : u_{B_1}(x) < \hat{t}\} = \{|x| > \sqrt{1 - 2n\hat{t}}\} = \{|x| > 1 - n\hat{t} + O(\hat{t}^2)\},$$

an annular shell of thickness $\simeq n\hat{t} \simeq_n w^2$, of measure

$$|\{x \in B_1 : u_{B_1} < \hat{t}\}| \simeq_n \mathcal{H}^{n-1}(\partial B_1) \cdot n\hat{t} \simeq_n w^2.$$

This is the w^2 . It is elementary: escaping a tube of height $\simeq w^2$ forces peeling a ball-shell of the same height, whose n -volume is $\simeq w^2$ *for every dimension*, irrespective of the (possibly tiny) tube volume.

(c) *The deficit is only $\simeq_n |T|$* . Energy expansion: attaching the thin tube changes $T(\Omega)$ by $\int_T u + (\text{junction}) \simeq w^2 |T| = O(w^{n+1} L)$ (negligible), while the equal-volume comparison ball B_* grows: $T(B_*) = T(B_1)(1 + |T|/\omega_n)^{(n+2)/n} = T(B_1)(1 + \frac{n+2}{n} \frac{|T|}{\omega_n} + \dots)$. Hence $T(B_*) - T(\Omega) \simeq_n |T|$ and

$$\delta_T(\Omega) \simeq_n |T| = \kappa_n w^{n-1} L \quad (\text{no } w^2 \text{ floor}).$$

(Independently confirmed by a direct level-set energy computation: $\delta_T = (2n^2 \omega_n)^{-1} |T| (1 + o(1))$.)

(d) *The obstruction to Strategy A.* The escape layer is $\lambda_{\text{esc}} \simeq_n |T| + w^2$, while the route’s budget (Lemma 2.2(ii)) is $|\Omega \setminus E_{\hat{t}}| = O(\delta_T) = O(|T|)$. For $n \geq 3$, $L = O(1)$, $w \rightarrow 0$ the shell dominates, $w^2 \gg w^{n-1} L = |T| \simeq \delta_T$, so $\lambda_{\text{esc}}/\delta_T \simeq w^{-(n-3)} \rightarrow \infty$: a tentacle-free (graph-like) level is *unreachable within an $O(\delta_T)$ layer for $n \geq 3$* . Only $n = 2$ survives ($|T| = \kappa_1 w L \gg w^2$ since $L \gg w$). This is exactly “tentacles yield non-graphs”.

3.4. Why this is not fatal to the program. The “weird” feeling is justified: the obstruction is to the *graph formulation only*, and the offending configuration is excluded by the very hypothesis $D_H(\hat{t}) \leq \theta$. A genuine tentacle has $|\nabla u| \simeq w \rightarrow 0$ on the tube part of Σ_t for $t < t_{\text{cap}}$. But by (6) D_H is the $1/|\nabla u|$ -weighted variance: on the tube part $|\nabla u| \simeq w$ is far from the mean $\bar{f} \simeq 1$ set by the dominant ball part, so the integrand is $(|\nabla u| - \bar{f})^2/|\nabla u| \simeq 1/w$. Explicitly, decompose the level

$\Sigma_t = \Gamma_{\text{ball}} \sqcup \Gamma_{\text{tube}}$ for $t < t_{\text{cap}}$: on Γ_{ball} , $|\nabla u| \simeq_n 1$ and $\mathcal{H}^{n-1}(\Gamma_{\text{ball}}) \simeq_n 1$; on Γ_{tube} , $|\nabla u| \simeq w$ and $\mathcal{H}^{n-1}(\Gamma_{\text{tube}}) \simeq_n w^{n-2}L$. The two-block Cauchy–Schwarz deficit is exactly

$$D_H(t) = A_0 A_1 \frac{(g_0 - g_1)^2}{g_0 g_1}, \quad A_i = \mathcal{H}^{n-1}(\Gamma_i), \quad g_i = \text{mean of } |\nabla u| \text{ on } \Gamma_i,$$

(the value of $(\int |\nabla u|^{-1})(\int |\nabla u|) - P^2$ for a two-valued $|\nabla u|$). With $g_0 \simeq 1$, $g_1 \simeq w$, $A_0 \simeq 1$, $A_1 \simeq w^{n-2}L$:

$$D_H(t) \simeq_n 1 \cdot w^{n-2}L \cdot \frac{1}{w} = w^{n-3}L$$

on a t -window of length $\simeq w^2$ (reproduced exactly by Lemma 4.4). The dimensional behaviour of $w^{n-3}L$ (thin tube, $w \rightarrow 0$, $L = O(1)$) is decisive and must be read correctly:

- $n = 2$: $D_H \simeq w^{-1}L \rightarrow \infty$ — a level with $D_H \leq \theta$ *cannot* pass through a tentacle; the tentacle is genuinely priced out.
- $n = 3$: $D_H \simeq L = O(1)$ — bounded; excluded only if the absolute constant exceeds θ , i.e. a constant-dependent borderline, not a robust exclusion.
- $n \geq 4$: $D_H \simeq w^{n-3}L \rightarrow 0$ — the tentacle has *small* D_H and therefore *satisfies* the near-Serrin hypothesis $D_H \leq \theta$. It is **not** excluded.

Correction of an earlier over-claim. A previous version of this note asserted “ $D_H \leq \theta$ forbids a tentacled level for $n \geq 3$, so the obstruction is vacuous”. That is **false for** $n \geq 4$: $w^{n-3}L \rightarrow 0$. For $n \geq 4$ a thin tentacle is a genuine near-Serrin level that is not a graph and cannot be escaped within an $O(\delta_T)$ layer — Strategy A is *genuinely obstructed* for $n \geq 4$, not vacuously. What survives is *Strategy B*, which never needs to exclude the tentacle: a tentacled level still has $\mathcal{A} \simeq |T|$ and $\delta_T \simeq |T|$, so $\mathcal{A}^2 \simeq |T|^2 \ll |T| \simeq \delta_T$ and the interior estimate holds with room. (The first agentic $L^2 \rightarrow L^\infty$ attempt did err by treating D_H as the *unweighted* variance; but the corrected weighted reading prices out tentacles only for $n = 2$, not for $n \geq 3$.)

Checkpoint (A1, the one you doubt — revised). The chain (a)–(d): $t_{\text{cap}} \simeq w^2$ (transverse barrier dominates, L irrelevant); ball-shell measure $\simeq_n w^2$; $\delta_T \simeq_n |T|$ (no w^2 floor in the deficit); hence the $n \geq 3$ graph obstruction. The $D_H \simeq w^{n-3}L$ tube-level scaling excludes tentacles *only for* $n = 2$; for $n \geq 4$ tentacles satisfy $D_H \leq \theta$, so Strategy A’s obstruction is *real and not vacuous* for $n \geq 4$. Strategy A is therefore genuinely a graph obstruction in dimension ≥ 4 ; only the interior route (Strategy B) survives it.

3.5. Honest status of Strategy A. Even granting (NDg) (delivered, in measure, by D_H via Lemma 4.4), two genuine difficulties remain. First, the available sharp nearly-spherical *torsional* inequality (Brasco–De Philippis–Velichkov) is, as literally proved, conditional on $\|\varphi\|_{C^{2,\gamma}}$ small, whereas the torsional deficit controls only $\|\varphi\|_{H^{1/2}}$ (see §5); the regularity-free $W^{1,\infty}$ torsional analogue is not in the corpus. Second, (Mod): even with (NDg), the interpolation needs a δ_T -uniform C^α/D^2u bound on $\Sigma_{\hat{t}}$, while interior Schauder gives only a constant $\sim \text{dist}(\Sigma_{\hat{t}}, \partial\Omega)^{-\beta}$ that degrades as the level nears a rough $\partial\Omega$.

Checkpoint (A2). Whether the perturbative theorem runs with the weak norm the deficit controls ($H^{1/2}$, not $C^{2,\gamma}$), and whether (Mod) can be made δ_T -uniform. *Open for Strategy A; precisely what Strategy B circumvents.*

4. STRATEGY B: A DIRECT INTERIOR APPROACH VIA WEINBERGER RIGIDITY

Serrin’s theorem states that if the torsion function of a domain has constant gradient on the boundary, the domain is a ball. Weinberger’s proof of this is not geometric: it produces an auxiliary function whose failure to be harmonic is exactly the trace-free part of D^2u , and that quantity vanishes precisely on the ball quadratic. Strategy B makes this rigidity *quantitative* and applies it, not to the rough set Ω , but to a single superlevel set of u , which is automatically smooth. Because the whole argument is interior, it uses *no* regularity of $\partial\Omega$ and never represents a level set as a graph.

4.1. The Weinberger function (classical). Let u solve (1) on a domain D . Set $P := |\nabla u|^2 + \frac{2}{n}u$. Since $\Delta u \equiv -1$,

$$\Delta P = \Delta |\nabla u|^2 + \frac{2}{n} \Delta u = 2|D^2 u|^2 - \frac{2}{n} = 2\left(|D^2 u|^2 - \frac{1}{n}\right).$$

The pointwise Cauchy–Schwarz inequality $|D^2 u|^2 \geq (\operatorname{tr} D^2 u)^2/n = (\Delta u)^2/n = 1/n$ gives $\Delta P \geq 0$, with equality at a point iff $D^2 u = -\frac{1}{n}I$ there, i.e. iff u agrees to second order with a ball quadratic $a - |x - x_0|^2/(2n)$. Writing the trace-free Hessian $(D^2 u + \frac{1}{n}I)$, one has $|(D^2 u + \frac{1}{n}I)|^2 = |D^2 u|^2 - \frac{1}{n} \geq 0$ and the exact identity

$$(7) \quad \Delta P = 2 \left| \left(D^2 u + \frac{1}{n} I \right) \right|^2.$$

Thus $\mathcal{R}(D) := \int_D |(D^2 u + \frac{1}{n}I)|^2$ measures how far u is from a ball quadratic; it vanishes iff D is a ball. Serrin/Weinberger rigidity is the qualitative statement “ $\mathcal{R}(D) = 0 \Rightarrow D$ a ball”. Strategy B is its quantitative form. This subsection is classical and not in question.

4.2. Step 1: reduction to one level set. By the common skeleton (Lemmas 2.2–2.4) it suffices to prove the sharp inequality on a single superlevel set. Precisely, there is a level $\hat{t}$ such that, with $\tilde{\Omega} := E_{\hat{t}} = \{u > \hat{t}\}$ and $\Sigma_{\hat{t}} = \{u = \hat{t}\}$: (i) $\Sigma_{\hat{t}}$ is a compact real-analytic hypersurface with $|\nabla u| > 0$ on it (Sard’s theorem and interior analyticity; no boundary regularity); (ii) the boundary defect is a small absolute constant, $\int_{\Sigma_{\hat{t}}} (|\nabla u| - \bar{f})^2 / |\nabla u| d\mathcal{H}^{n-1} \leq \theta$, where $\bar{f}$ is the mean of $|\nabla u|$ on $\Sigma_{\hat{t}}$; (iii) $|\Omega \setminus \tilde{\Omega}| \leq C_n \delta_T(\Omega)/\theta$; and (iv) $\delta_T(\tilde{\Omega}) \leq 2\delta_T(\Omega)$. Moreover $u - \hat{t}$ is *exactly* the torsion function of $\tilde{\Omega}$. So it suffices to prove $\mathcal{A}(\tilde{\Omega})^2 \leq C(n) \delta_T(\tilde{\Omega})$, which the next two steps establish.

4.3. Step 2: the deficit dominates the Hessian defect.

Proposition 4.1. *There is a dimensional constant $c(n) > 0$ such that, for every bounded domain D with the regularity of a Sard-regular level set (in particular $D = \tilde{\Omega}$),*

$$\delta_T(D) \geq c(n) |D|^{-1} \mathcal{R}(D).$$

The constant depends only on n — not on any regularity, inradius, or shape constant of D — and is additive over connected components.

Idea. Two classical identities for v the torsion function of D ($v = 0$, $\nabla v = v_\nu \nu$ on ∂D). First, the Rellich–Pohozaev identity, obtained by testing $-\Delta v = 1$ with $x \cdot \nabla v$ and integrating by parts:

$$\frac{1}{2} \int_{\partial D} |\nabla v|^2 (x \cdot \nu) d\mathcal{H}^{n-1} = \frac{n+2}{2} \int_D v = \frac{n+2}{2} T(D).$$

Second, integrating (7) and using the boundary value $P = |\nabla v|^2$ on ∂D writes the Saint–Venant gap $T(B_D) - T(D)$ both as a nonnegative boundary functional of $|\nabla v|$ on ∂D and, pairing (7) with the Green function of D (a positive weight requiring no boundary regularity), as a dimensional multiple of $\int_D |(D^2 u + \frac{1}{n}I)|^2 = \mathcal{R}(D)$. The pointwise inequality $|D^2 u|^2 \geq 1/n$, sharp only on the ball quadratic, makes the two expressions comparable. Combining them yields the stated inequality with $c(n)$ dimensional. This is precisely the quantitative form of Weinberger’s rigidity. $\square$

Checkpoint (B1). That Proposition 4.1 holds with a *purely dimensional* constant, using only that the Rellich–Pohozaev and Green-function identities are valid on a Sard-regular interior-analytic level (no $\partial\Omega$ regularity), and that it is stable under disconnection and thin tentacles. Consistency check: on a ball-with-thin-tube model both sides scale like the tube volume.

4.4. Step 3: the Hessian defect dominates the asymmetry. The engine is a rigidity for the Hessian.

Lemma 4.2 (quantitative Liouville for the Hessian). *Let $D \subset \mathbb{R}^n$ be open with $|D| = \omega_n$ and $w \in H^2(D)$. There exist a quadratic $q(x) = a - |x - x_0|^2/(2n)$ and a constant $K(D)$ with*

$$\int_D |\nabla w - \nabla q|^2 \leq K(D) \int_D |D^2 w + \frac{1}{n}I|^2.$$

$K(D)$ is a Poincaré-type constant of D ; it is dimensional for D in a controlled class. Controlling $K(D)$ for the level domains that arise is the load-bearing analytic point of Strategy B.

Proposition 4.3. *With $D = \tilde{\Omega}$ there is a dimensional $C(n)$ with $\mathcal{A}(D)^2 \leq C(n) |D|^{-1} \mathcal{R}(D)$.*

Idea. Apply Lemma 4.2 with $w = u$: if $|D|^{-1} \mathcal{R}(D)$ is small then $D^2 u \approx -\frac{1}{n} I$ in L^2 , so u is H^2 -close to a quadratic q ; its superlevel set $\{u > \hat{t}\}$ is then close in measure to the ball $\{q > \hat{t}\}$ at the sharp (quadratic) rate, and the Fraenkel asymmetry, being the squared L^2 size of the boundary deviation, is controlled by $|D|^{-1} \mathcal{R}(D)$. The one delicate point is that $K(D)$ in Lemma 4.2 could degrade where ∇u is small (the level set can pinch there). Remove the set $Z = \{|\nabla u| \leq s_0\}$ at a dimensional threshold $s_0 \simeq_n 1/n$ and bound its measure directly:

Lemma 4.4 (the small-gradient set is small). *On $\Sigma_{\hat{t}}$, for every $s > 0$, $\mathcal{H}^{n-1}(\{|\nabla u| \leq s\} \cap \Sigma_{\hat{t}}) \leq 4s \bar{f}^{-2} \int_{\Sigma_{\hat{t}}} (|\nabla u| - \bar{f})^2 / |\nabla u| d\mathcal{H}^{n-1}$.*

(Where $|\nabla u| \leq s \leq \bar{f}/2$ one has $(|\nabla u| - \bar{f})^2 / |\nabla u| \geq \bar{f}^2 / (4s)$; integrate over that set only — Chebyshev applied to the defect integral of (ii).) By (ii) the right side is a small absolute constant, so $|Z|$ is small and is paid for *additively*: removing a piece changes the asymmetry by at most its relative measure. On the remaining set $D \setminus Z$ one has $|\nabla u| \geq s_0$, so $K(D \setminus Z)$ is dimensional, and the divergent contribution that would come from the pinch region has been excised rather than multiplied against surviving terms. If D is disconnected the argument runs per component; a component of order-one asymmetry has order-one deficit by Lemma 4.5, hence lies outside the small-deficit regime, so the per-component treatment loses nothing. $\square$

Lemma 4.5 (strict-stability threshold). *There are $\eta_0(n), c_0(n) > 0$ with: any open D , $|D| = \omega_n$, $\mathcal{A}(D) \geq \eta_0$ satisfies $\delta_T(D) \geq c_0$. (δ_T is lower semicontinuous and positive off balls on the compact-modulo-translation set $\{\mathcal{A} \geq \eta_0\}$.)*

Checkpoint (B2, decisive). That $K(D \setminus Z)$ in Lemma 4.2 is *dimensional* uniformly over the admissible level domains — possibly pinched or multi-component — once the small-gradient set Z is excised. Concretely, on a ball-with-tube model the true values are $\mathcal{A} \simeq |T|$ and $\mathcal{R} \simeq |T|$, so Proposition 4.3 must give $\mathcal{A}^2 \simeq |T|^2 \leq C |T|$ with C dimensional and room to spare. Everything else in Step 3 is elementary or is Lemma 4.4; *this is the single point on which the strategy rests.*

4.5. Step 4: conclusion.

Theorem 4.6. (Conditional on Checkpoints B1, B2.) *There is $C_*(n)$ such that every finite-measure open $\Omega \subset \mathbb{R}^n$ with $\delta_T(\Omega) \leq \delta_T^0(n)$ satisfies $\mathcal{A}(\Omega)^2 \leq C_*(n) \delta_T(\Omega)$, sharp exponent.*

Proof. On $\tilde{\Omega} = E_{\hat{t}}$, Propositions 4.3 and 4.1 give $\mathcal{A}(\tilde{\Omega})^2 \leq C(n) |\tilde{\Omega}|^{-1} \mathcal{R}(\tilde{\Omega}) \leq (C(n)/c(n)) \delta_T(\tilde{\Omega})$, and $\delta_T(\tilde{\Omega}) \leq 2\delta_T(\Omega)$ by Step 1(iv). Lemma 2.4 transfers: $\mathcal{A}(\Omega)^2 \leq 2\mathcal{A}(\tilde{\Omega})^2 + 8(|\Omega \setminus \tilde{\Omega}|/|\Omega|)^2 \leq C_*(n) \delta_T(\Omega)$, the squared collar being $O(\delta_T^2) \ll \delta_T$ by Step 1(iii). No boundary regularity, no graph, and no perturbative nearly-spherical step is used. $\square$

5. SHARPNESS AND THE NORM GAP

Why does Strategy A stall where B does not? Expand a nearly-spherical $\partial D = \{(1 + \varphi(\xi))\xi : \xi \in \mathbb{S}^{n-1}\}$ in spherical harmonics $\varphi = \sum_{k \geq 0} \sum_m \hat{\varphi}_{k,m} Y_{k,m}$, $-\Delta_{\mathbb{S}^{n-1}} Y_{k,m} = k(k+n-2) Y_{k,m}$. The volume constraint $|D| = \omega_n$ pins the mode $k = 0$ at second order ($\hat{\varphi}_0 = -\frac{1}{2} \sum_{k \geq 1} |\hat{\varphi}_k|^2 / \dots$), and choosing the optimal centre kills the modes $k = 1$ (translations are asymmetry-neutral). The Fuglede second variation of the torsional energy is

$$\delta_T(D) \simeq_n \sum_{k \geq 2} \gamma_k |\hat{\varphi}_k|^2, \quad \gamma_k = \frac{(k-1)(2k+n-2) + \dots}{k+n-2} \asymp k,$$

so the surviving quadratic form is comparable to $\sum_{k \geq 2} k |\hat{\varphi}_k|^2 \simeq \|\varphi\|_{H^{1/2}(\mathbb{S}^{n-1})}^2$ (the Steklov/Dirichlet-to-Neumann scale), *not* $\sum k^2 |\hat{\varphi}_k|^2 \simeq \|\varphi\|_{H^1}^2$. The gap $\gamma_k > 0$ for $k \geq 2$ is the quantitative Saint-Venant

rigidity; its *linear* (not quadratic) growth in k is the precise reason the deficit controls only a half-derivative of the graph. There is an explicit family Ω_h with $\|\varphi_h\|_{H^{1/2}}$ bounded but $\|\varphi_h\|_{H^1} \rightarrow \infty$: *the deficit cannot control more than $H^{1/2}$ of the graph*. The BDV nearly-spherical theorem, as written, needs $\|\varphi\|_{C^{2,\gamma}}$ small (its Hadamard-flow remainder, not its $H^{1/2}$ Steklov core, forces the two extra derivatives). So Strategy A must either supply $C^{2,\gamma}$ smallness regularity-free (the dead (BReg) route) or reprove BDV in the flow-free $H^{1/2}$ norm. Strategy B sidesteps this entirely: Propositions 4.1 and 4.3 never see a graph norm; they trade through the interior H^2 quantity $\mathcal{R}$, for which the sharp quadratic relation is intrinsic (the $|D^2u|^2 \geq 1/n$ Newton inequality is exactly order 2).

6. THE FABER–KRAHN EIGENVALUE VERSION

The title says Faber–Krahn; we record how the torsional statement (2) delivers the eigenvalue one. Let $\lambda_1(\Omega)$ be the first Dirichlet eigenvalue, $\lambda_1(B)$ its ball value at equal volume, and $\delta_\lambda(\Omega) = (\lambda_1(\Omega) - \lambda_1(B))/\lambda_1(B)$ the Faber–Krahn deficit. Two routes connect the two functionals.

- *Kohler-Jobin*. The Kohler-Jobin inequality $T(\Omega) \lambda_1(\Omega)^{(n+2)/2} \geq T(B) \lambda_1(B)^{(n+2)/2}$, sharp on balls, gives $\delta_T(\Omega) \leq c(n) \delta_\lambda(\Omega)$ to first order: a small eigenvalue deficit forces a small torsional deficit, so (2) yields $\mathcal{A}(\Omega)^2 \leq C(n) \delta_\lambda(\Omega)$ — the sharp quantitative Faber–Krahn inequality — with no extra work.
- *Direct analogue*. The first eigenfunction $u_1 > 0$ solves $-\Delta u_1 = \lambda_1 u_1$; the Weinberger P -function for the eigenvalue problem, $P_\lambda = |\nabla u_1|^2 + \lambda_1 u_1^2$, is again subharmonic with $\Delta P_\lambda = 2|D^2u_1 + \frac{\lambda_1}{n}u_1 I|^2 \geq 0$ and the same trace-free Hessian rigidity. Strategy B transfers verbatim with $\mathcal{R}$ replaced by $\int |D^2u_1 + \frac{\lambda_1}{n}u_1 I|^2$ and the level sets of u_1 in place of those of u ; the level set $\{u_1 > \hat{t}\}$ has its own eigenfunction $u_1 - \hat{t}$ only after the standard substitution, so the cleaner statement goes through Kohler-Jobin. Either way the eigenvalue version is a corollary, not a separate problem.

Checkpoint (FK). The Kohler-Jobin reduction constant and its sharpness on balls (classical; see Brasco–De Philippis–Velichkov). Only needed if the eigenvalue statement is wanted directly rather than as a corollary of (2).

7. COMPARISON WITH PLAN 1, AND WHAT TO TRUST

7.1. Plan 1 vs. the two strategies. Plan 1 (the BDV *selection principle*) replaces an arbitrary near-extremal Ω by a penalised minimiser $\tilde{\Omega}$ that is, by free-boundary regularity, a smooth nearly-spherical set, with deficit not worse and asymmetry not smaller, then closes by the nearly-spherical inequality. Its hard core is the selection step’s regularity (the $C^{1,\gamma}$ /free-boundary constant tracking, Plan 1’s Lemmas 6–7). *Strategy A is the literal analogue* — the smooth competitor is a torsion level set, Serrin stability the regulariser — and inherits both a regularity gap and the $n \geq 3$ tentacle/ w^2 obstruction to graphicality. *Strategy B replaces the selection principle outright*: it never produces a smooth competitor; it applies quantitative Weinberger directly to the level set, so no graph, no $C^{2,\gamma}$, no selection principle — the regularity-free small-gradient bound (Lemma 4.4, driven by D_H) supplies the only non-degeneracy the argument needs. Strategy B is the realisation of “do Plan 1’s idea, but with Serrin, directly”.

7.2. Consolidated checklist. Common skeleton. C1 (Lemma 2.2); C2 (Lemma 2.3, profile shift & sign); C3 (Lemma 2.4 & rescaling composition).

Strategy A. A1 (§3.3: $t_{\text{cap}} \simeq w^2$; ball-shell $\simeq_n w^2$; $\delta_T \simeq_n |T|$; $n \geq 3$ obstruction; the $D_H \simeq w^{n-3}L$ tube scaling excludes tentacles *only for $n = 2$ — the w^2 you asked about, derived in §3.3(b)*; for $n \geq 4$ tentacles satisfy $D_H \leq \theta$ and the graph obstruction is genuine). A2 (perturbative theorem in the weak norm $H^{1/2}$; δ_T -uniform (Mod)) — *open*; B *circumvents it*.

Strategy B (decisive). B1 (Prop. 4.1: regularity-free quantitative Weinberger, dimensional constant, stable under disconnection/tentacles). B2 (Prop. 4.3 via Lemma 4.2: H^2 -rigidity $\Rightarrow$ sharp Fraenkel,

dimensional Poincaré constant on pinched/disconnected D after excising the small-gradient set — *the single most important line*). B3 (Lemma 4.4 & its agreement with the tentacle D_H scaling).

7.3. What is, and is not, established. Lemmas 2.2–2.4 are routine (C1–C3 bookkeeping). Strategy A’s $n \geq 3$ obstruction is, modulo A1, correct arithmetic about tentacled domains; the $D_H \simeq w^{n-3}L$ scaling prices tentacles out only for $n = 2$, so for $n \geq 4$ the obstruction is *genuine, not vacuous* — a thin tentacle is a bona fide near-Serrin level that is not a graph. It remains irrelevant to Strategy B, which never excludes the tentacle: such a level still has $\mathcal{A}^2 \simeq |T|^2 \ll |T| \simeq \delta_T$, so the interior estimate holds with room. Strategy B’s analytic inputs B1, B2 have been derived and passed an internal adversarial verification with an independent cross-check (B3 reproduces the tentacle scaling exactly), but are **not** human-refereed. Honest status: the program plausibly closes (2) for all n via Strategy B, conditional on B1 and especially B2 surviving a full referee; Strategy A, as a graph route, is genuinely obstructed for $n \geq 3$ and is the heuristic motivating B, not a proof.